

These are the commands for the follow-along exercises in Lesson 3.5.

3.502–3.508: Refactoring with Django Generic Views Parts A–G

views.py

```
from django.views.generic import ListView
```

```
class GeneList(ListView):
    model = Gene
    context_object_name = 'master_genes'
    template_name = 'genedata/index.html'
```

urls.py

```
path('', views.GeneList.as_view(), name='index'),
```

views.py

```
from django.views.generic import DetailView
```

```
class GeneDetail(DetailView):
    model = Gene
    context_object_name = 'gene'
    template_name = 'genedata/gene.html'
```

urls.py

```
path('gene/<int:pk>', views.GeneDetail.as_view(), name='gene'),
```

views.py

```
def get_context_data(self, **kwargs):
    context = super().get_context_data(**kwargs)
    context['master_genes'] = Gene.objects.all()
    return context
```

views.py

```
from django.views.generic.edit import CreateView
```

```
class GeneCreate(CreateView):
    model = Gene
    template_name = 'genedata/create_gene.html'
    form_class = GeneForm
    success_url = "/create_gene/"

    def get_context_data(self, **kwargs):
        context = super().get_context_data(**kwargs)
        context['master_genes'] = Gene.objects.all()
        return context
```

urls.py

```
path('create_gene/', views.GeneCreate.as_view(),
name='create_gene'),
```

views.py

```
from django.views.generic.edit import DeleteView
```

```
class GeneDelete(DeleteView):
```

```
    model = Gene
```

```
    success_url = "/"
```

```
    def get_context_data(self, **kwargs):
```

```
        context = super().get_context_data(**kwargs)
```

```
        context['master_genes'] = Gene.objects.all()
```

```
        return context
```

urls.py

```
path('delete/<int:pk>', views.GeneDelete.as_view(),
name='delete'),
```

gene_confirm_delete.html

```
{% block content %}
```

```
<form method="post">{% csrf_token %}
```

```
    <p>Are you sure you want to delete "{{ object }}"?</p>
```

```
    <input type="submit" value="Confirm">
```

```
</form>
```

```
{% endblock %}
```

gene.html

```
<a href="/update/{{gene.pk}}">UPDATE RECORD</a>      <br />
```

```
<a href="/delete/{{gene.pk}}">DELETE RECORD</a>
```

urls.py

```
path('update/<int:pk>', views.GeneUpdate.as_view(),
name='update'),
```

views.py

```
from django.views.generic.edit import UpdateView
```

```
class GeneUpdate(UpdateView):
```

```
    model = Gene
```

```
    fields = fields = ['gene_id', 'entity', 'start', 'stop',
'sense', 'start_codon',
                        'sequencing', 'ec']
```

```
    template_name_suffix = '_update_form'
```

```
    def get_context_data(self, **kwargs):
```

```
        context = super().get_context_data(**kwargs)
```

```
        context['master_genes'] = Gene.objects.all()
```

```
        return context
```

```
touch genedata/templates/genedata/gene_update_form.html
```

gene_update_form.html

```
{% extends "../base.html" %}

{% block content %}
<form method="post">{% csrf_token %}
    {{ form.as_p }}
    <input type="submit" value="Update">
</form>
{% endblock %}
```

urls.py

```
path('list/<str:type>', views.GeneList.as_view(),
name='list'),
```

views.py

```
class GeneList(ListView):
    model = Gene
    context_object_name = 'master_genes'

    def get_context_data(self, **kwargs):
        context = super().get_context_data(**kwargs)
        context['master_genes'] = Gene.objects.all()
        if 'type' in self.kwargs:
            if "Chromosome" in self.kwargs['type'] or "Plasmid" in
self.kwargs['type']:
                context['genes'] =
Gene.objects.filter(entity__exact=self.kwargs['type'])
            return context

        def get_template_names(self):
            if 'type' in self.kwargs:
                if "Chromosome" in self.kwargs['type'] or "Plasmid" in
self.kwargs['type']:
                    return 'genedata/list.html'

            return 'genedata/index.html'
```

urls.py

```
path('poslist/', views.GeneList.as_view(), name='poslist'),
```

views.py

```
GeneList(ListView):
    model = Gene
    context_object_name = 'master_genes'

    def get_context_data(self, **kwargs):
        context = super().get_context_data(**kwargs)
        context['master_genes'] = Gene.objects.all()
        if "poslist" in self.request.get_full_path():
```

```

        context['genes'] =
Gene.objects.filter(entity__exact='Chromosome').filter(sense__star
tswith='+')
        if 'type' in self.kwargs:
            if "Chromosome" in self.kwargs['type'] or "Plasmid" in
self.kwargs['type']:
                context['genes'] =
Gene.objects.filter(entity__exact=self.kwargs['type'])
            return context

def get_template_names(self):
    if "poslist" in self.request.get_full_path():
        return 'genedata/list.html'
    if 'type' in self.kwargs:
        if "Chromosome" in self.kwargs['type'] or "Plasmid" in
self.kwargs['type']:
            return 'genedata/list.html'

    return 'genedata/index.html'

```